

Conjecture 1. Let $\tau(n)$ denote the number of positive divisors of n . No integer $n > 24$ satisfies

$$\max_{1 \leq m < n} (m + \tau(m)) \leq n + 2.$$

Moreover, $n = 24$ is the largest positive integer satisfying this inequality.

Theorem 1. Let $f(m) = m + \tau(m)$. For every integer $n > 24$ except possibly those of the form

$$n = 2520j, \quad 420j-1 \text{ prime}, \quad 504j-1 \text{ prime}, \quad j \equiv 0 \text{ or } 7 \pmod{11}, \quad j > 10^7,$$

there exists $m < n$ with $f(m) > n + 2$. The proof divides into three parts:

- (A) **Algebraic.** If $6 \nmid n$; if $6 \mid n$ and $\tau(n-6) \neq 8$; or if $n = 6(p+1)$ with p prime and $p \not\equiv 419 \pmod{420}$: a witness exists unconditionally for every such $n > 24$.
- (B) **Covering.** If $n = 2520j$ with $420j-1$ prime and $j \not\equiv 0, 7 \pmod{11}$: a witness exists for every $j \geq 1$ via the mod-11 covering argument (Lemma 2).
- (C) **Computational.** If $n = 2520j$ with $420j-1$ and $504j-1$ both prime and $j \equiv 0 \text{ or } 7 \pmod{11}$: a witness exists for all $j \leq 10^7$ by exhaustive search (Claim 1).

We write $f(m) = m + \tau(m)$ throughout, so the condition states $f(m) \leq n + 2$ for all $m < n$. We prove Theorem 1 by exhibiting a witness $m < n$ with $f(m) > n + 2$.

Lemma 1. For every integer $t \geq 4$, $\tau(6t) \geq 8$.

Proof. Write $6t = 2^a \cdot 3^b \cdot r$ where r is a positive integer with $\gcd(r, 6) = 1$, $a \geq 1$, $b \geq 1$. Since $t \geq 4$, we have $2^{a-1} \cdot 3^{b-1} \cdot r \geq 4$. We check all shapes of t :

- $a \geq 3$: $\tau(6t) = (a+1)(b+1)\tau(r) \geq 4 \cdot 2 = 8$.
- $a = 2, b \geq 2$: $\tau(6t) = 3(b+1) \geq 9$.
- $a = 2, b = 1, r > 1$: $\tau(6t) = 3 \cdot 2 \cdot \tau(r) \geq 12$.
- $a = 1, b \geq 3$: $\tau(6t) = 2(b+1) \geq 8$.
- $a = 1, b = 2, r > 1$: $\tau(6t) = 2 \cdot 3 \cdot \tau(r) \geq 12$.

- $a = 1, b = 1, r > 1$: $\tau(6t) = 2 \cdot 2 \cdot \tau(r) \geq 8$.

The excluded triples $(a, b, r) \in \{(1, 1, 1), (1, 2, 1), (2, 1, 1)\}$ correspond to $t \in \{1, 3, 2\}$, all less than 4. In every remaining case $\tau(6t) \geq 8$. $\square$

Proof of Theorem 1. Part A (algebraic cases). Let $n > 24$. We split into three cases according to the residue of n modulo 6.

Case 1: n is odd. Set $m = n - 1$ (even, $m \geq 24$). We show $\tau(m) \geq 4$.

- $\tau(m) = 1$: impossible since $m \geq 24$.
- $\tau(m) = 2$: m is prime, but m is even and $m > 2$, impossible.
- $\tau(m) = 3$: $m = q^2$ for some prime q . Since m is even, $q = 2$, giving $m = 4, n = 5$, contradicting $n > 24$.

Hence $\tau(m) \geq 4$ and $f(m) = (n - 1) + \tau(m) \geq n + 3 > n + 2$.

Case 2: n is even and $6 \nmid n$. Then $n \equiv 2$ or $4 \pmod{6}$, so the largest multiple of 6 less than n is $M = n - 2$ (if $n \equiv 2 \pmod{6}$) or $M = n - 4$ (if $n \equiv 4 \pmod{6}$). Set $t = M/6$. Since $n \geq 26$, we have $t = M/6 \geq (n-4)/6 \geq 22/6 > 3$, so $t \geq 4$. By Lemma 1, $\tau(M) \geq 8$. Since $n - M \in \{2, 4\}$,

$$f(M) = M + \tau(M) \geq M + 8 = n - (n - M) + 8 \geq n - 4 + 8 = n + 4 > n + 2.$$

Case 3: $6 \mid n, n \geq 30$. Write $n = 6s$ with $s \geq 5$. Set $M = n - 6 = 6(s - 1)$, so $t = s - 1 \geq 4$. By Lemma 1, $\tau(M) \geq 8$.

Sub-case 3a: $\tau(M) \geq 9$. Then $f(M) = n - 6 + \tau(M) \geq n + 3 > n + 2$.

Sub-case 3b: $\tau(M) = 8$. Here $f(M) = n + 2$, which is not strictly greater.

We look for a different witness near n .

Claim: At least one of $m \in \{n - 1, n - 3, n - 4\}$ satisfies $f(m) > n + 2$.

The needed divisibility conditions are:

$$\begin{aligned} m = n - 1 : \quad & \tau(n - 1) \geq 4, \\ m = n - 3 : \quad & \tau(n - 3) \geq 6, \\ m = n - 4 : \quad & \tau(n - 4) \geq 7. \end{aligned}$$

We carry out the complete case analysis below.

Structure of Sub-case 3b. Here $\tau(n - 6) = 8$. Since $n - 6 = 6(s - 1)$ and $s - 1 \geq 4$, the number $6(s - 1)$ has exactly 8 divisors in precisely three shapes:

- $6(s-1) = 2^3 \cdot 3 = 24$, giving $n = 30$;
- $6(s-1) = 2 \cdot 3^3 = 54$, giving $n = 60$;
- $6(s-1) = 2 \cdot 3 \cdot p = 6p$ with $p \geq 5$ prime, giving $n = 6(p+1)$.

The first two are handled directly:

- $n = 30$: $m = 28 = 4 \cdot 7$, $\tau(28) = 6$, $f(28) = 34 > 32 = 30 + 2$.
- $n = 60$: $m = 56 = 2^3 \cdot 7$, $\tau(56) = 8$, $f(56) = 64 > 62 = 60 + 2$.

For the third shape, $n = 6(p+1)$ with $p \geq 5$ prime. We set $n = 6p + 6$ throughout.

We check candidates $m \in \{n-1, n-2, n-3, n-4, n-5\}$ in turn. The witness condition $f(m) > n+2$ is $\tau(n-k) > k+2$ for $m = n-k$.

(i) $p \equiv 1 \pmod{3}$. Then $3 \mid (2p+1)$ and in fact $9 \mid n-3$ because $n-3 = 3(2p+1)$. Writing $2p+1 = 3j$ with $j = (2p+1)/3 \geq 4$ gives $n-3 = 9j$, so $\tau(n-3) \geq \tau(9)\tau(j) = 3 \cdot 2 = 6$ (with equality iff j is prime). Hence $f(n-3) \geq n+3 > n+2$.

(ii) $p \equiv 2 \pmod{3}$, $p \equiv 1 \pmod{5}$. Then $5 \mid (3p+2)$, so $3p+2 \geq 5$ and $3p+2 = 5j$ with $j \geq 4$. Since $n-2 = 2(3p+2) = 10j$ and $\gcd(2, j) \in \{1, 2\}$: $\tau(n-2) \geq \tau(10)\tau(j/\gcd(2, j)) \cdot \gcd(2, j)/1 \geq 8$. Hence $f(n-2) = n-2 + \tau(n-2) \geq n+6 > n+2$.

(iii) $p \equiv 2 \pmod{3}$, $p \equiv 2 \pmod{5}$. Then $5 \mid (2p+1)$, so $n-3 = 3(2p+1) = 15j$ with $j = (2p+1)/5 \geq 5$. $\tau(n-3) \geq \tau(15)\tau(j) \geq 4 \cdot 2 = 8$. Hence $f(n-3) \geq n+5 > n+2$.

(iv) $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{5}$. By CRT, $p \equiv 23$ or $53 \pmod{60}$. Since $p \equiv 3 \pmod{5}$: $5 \mid (3p+1)$. Since p is odd: $2 \mid (3p+1)$. We split on $p \pmod{4}$:

- $p \equiv 1 \pmod{4}$ (so $p \equiv 53 \pmod{60}$, smallest case $p = 53$): $3p \equiv 3 \pmod{4}$, so $4 \mid (3p+1)$. Combined with $5 \mid (3p+1)$: $20 \mid (3p+1)$. Write $3p+1 = 20w$; since $p \geq 53$, $w \geq 8$. Then $n-4 = 40w$. Since $40 \mid 40w$, every divisor of 40 is a divisor of $40w$, so $\tau(n-4) \geq \tau(40) = 8$ and $f(n-4) \geq n+4 > n+2$.
- $p \equiv 3 \pmod{4}$ (so $p \equiv 23 \pmod{60}$, smallest case $p = 23$): $3p \equiv 9 \equiv 1 \pmod{4}$, so $v_2(3p+1) = 1$. Combined with $5 \mid (3p+1)$, write $3p+1 = 10t$ with t odd; since $p \geq 23$, $t = (3p+1)/10 \geq 7$. Then $n-4 = 20t$ with $t \geq 7$ odd. If $\gcd(t, 5) = 1$: since t is odd, $\gcd(t, 20) = 1$, so $\tau(20t) = \tau(4)\tau(5)\tau(t) = 6\tau(t) \geq 12$. If $5 \mid t$: write $t = 5^a s$ with $5 \nmid s$, $a \geq 1$; then $\tau(20t) = \tau(4)\tau(5^{a+1})\tau(s) = 3(a+2)\tau(s) \geq 9$. In either case $\tau(n-4) \geq 9$ and $f(n-4) \geq n+5 > n+2$.

(v) $p \equiv 2 \pmod{3}$, $p \equiv 4 \pmod{5}$. Then $5 \mid (6p+1)$ (since $6p+1 \equiv 24+1=25 \equiv 0 \pmod{5}$), so $n-5=6p+1=5s$ with $s=(6p+1)/5$.

- If s is composite: $\tau(s) \geq 4$, $\tau(n-5) = \tau(5s) \geq 8$. $f(n-5) \geq n+3 > n+2$.

- If s is prime: then $p \equiv 4 \pmod{5}$ and $p \equiv 2 \pmod{3}$. Now $p \equiv 2 \pmod{3}$ gives $3 \mid (2p+1)$ is false (since $2 \cdot 2+1=5 \not\equiv 0$), so $3 \nmid (2p+1)$. Consider sub-cases by $p \pmod{7}$:

- $p \equiv 3 \pmod{7}$: $7 \mid (2p+1)$, so $\tau(n-3) \geq \tau(3 \cdot 7 \cdot (2p+1)/21) \cdot 4 \geq 8$.
- $p \equiv 4 \pmod{7}$: $7 \mid (3p+2)$, $\tau(n-2) \geq 8$.
- $p \equiv 5 \pmod{7}$: $7 \mid (6p+5) = n-1$, so $n-1$ is composite and $\tau(n-1) \geq 4$, giving $f(n-1) > n+2$.
- $p \equiv 2 \pmod{7}$: $7 \mid (3p+1)$ (since $3 \cdot 2+1=7 \equiv 0 \pmod{7}$). We split on $p \pmod{4}$:
 - * $p \equiv 1 \pmod{4}$: $3p+1 \equiv 3+1=4 \equiv 0 \pmod{4}$, so $28 \mid (3p+1)$ and $56 \mid n-4=2(3p+1)$. Hence $\tau(n-4) \geq \tau(56) = 8 > 6$.
 - * $p \equiv 3 \pmod{4}$: $3p+1 \equiv 9+1=10 \equiv 2 \pmod{4}$, so $v_2(3p+1) = 1$. Write $n-4=4u$ with $u=(3p+1)/2$ odd and $7 \mid u$ (since $7 \mid 3p+1$ and $\gcd(7,2)=1$). The combined residue class is $p \equiv 359 \pmod{420}$, so $p \geq 359$ and $u=(3p+1)/2 \geq 539$; write $u=7v$ with $v \geq 77$. Since $v > 1$, $\tau(7v) \geq 4$ (as $7v$ is composite with both 7 and v as factors), so $\tau(n-4) = \tau(4u) = 3\tau(7v) \geq 12 > 6$.
- $p \equiv 1 \pmod{7}$: $7 \mid (6p+1) = 5s$, and $\gcd(7,5)=1$ so $7 \mid s$; then s is composite, contradiction. So this sub-sub-case cannot arise when s is prime.
- $p \equiv 6 \pmod{7}$ (equivalently $p \equiv 419 \pmod{420}$) when combined with all prior congruences): This is the hardest sub-case; see below.
- $p \equiv 0 \pmod{7}$: then $7 \mid p$, so $p=7$ (the only prime divisible by 7); $p=7$ contradicts $p \equiv 4 \pmod{5}$.

Part B(mod-11 covering). The sub-cases above handle all primes p except $p \equiv 419 \pmod{420}$. The argument here combines direct congruence analysis (to produce A_k divisible by 11) with a divisor-count lower bound; the method is not a single unified technique but an exhaustive case split that happens to close under one prime-modulus argument.

The hardest sub-case: $p \equiv 419 \pmod{420}$. Here $p \equiv 59 \pmod{60}$, $p \equiv 6 \pmod{7}$, and $(6p+1)/5$ is prime. Write $p = 420j - 1$ for a positive integer j . Since $p+1 = 420j$,

$$n = 6(p+1) = 2520j.$$

Note that $(6p+1)/5 = (6(420j-1)+1)/5 = (2520j-5)/5 = 504j-1$, so the primality condition on $(6p+1)/5$ is exactly the condition $504j-1$ prime that appears in Theorem 1 and Proposition 1. For any k with $k \mid 2520$, set $A_k = \frac{2520}{k}j - 1$, so that

$$n - k = k A_k.$$

The witness condition $\tau(n - k) > k + 2$ becomes $\tau(k A_k) > k + 2$.

Lemma 2 (Algebraic Covering). *For every positive integer j with $j \not\equiv 0$ or $7 \pmod{11}$, there exists $k \in \{1, 2, 3, 4, 5, 7, 8, 9, 10\}$ with $k \mid 2520$ such that $\tau(k A_k) > k + 2$.*

Proof. Since $2520 \equiv 1 \pmod{11}$, for any $k \mid 2520$ with $k \leq 10$ we have $\frac{2520}{k} \cdot k = 2520 \equiv 1 \pmod{11}$, hence

$$11 \mid A_k \iff j \equiv k \pmod{11}.$$

We treat the nine residue classes $j \not\equiv 0, 7 \pmod{11}$ separately.

Cases $j \equiv r \pmod{11}$, $r \in \{1, 2, 3, 4, 5, 8, 9, 10\}$ (algebraic cases).

Use $k = r$. Then $11 \mid A_r$, so $A_r = 11M$ for some positive integer M . Since $A_r = \frac{2520}{r}j - 1 \geq \frac{2520}{r} \cdot r - 1 = 2519 = 11 \cdot 229$, we have $M \geq 229 > 1$.

We need $\tau(r \cdot A_r) > r + 2$. Write $A_r = 11M$. Since A_r is odd (as $\frac{2520}{r}j$ is even) and $11 \nmid r$ (as $r \leq 10$), the integers r , 11 , and M' (where $A_r = 11^{v_{11}(A_r)} \cdot M'$, $11 \nmid M'$) satisfy:

$$\tau(r \cdot A_r) \geq \tau(r) \cdot \tau(A_r) / \tau(\gcd(r, A_r)) \geq \tau(r) \cdot \tau(11M).$$

Since $M \geq 229 > 1$, we have $\tau(11M) \geq 4$ (regardless of whether $11 \mid M$ or not: if $11 \parallel A_r$ then $\tau(11M) = 2\tau(M) \geq 4$; if $11^a \parallel A_r$ with $a \geq 2$ then $\tau(A_r) \geq 3\tau(M') \geq 6$). We verify $\tau(r) \cdot 4 > r + 2$ for each r :

r	$\tau(r)$	$\tau(r) \cdot 4$	$r + 2$
1	1	4	3
2	2	8	4
3	2	8	5
4	3	12	6
5	2	8	7
8	4	16	10
9	3	12	11
10	4	16	12

In every row $\tau(r) \cdot 4 > r + 2$, so $\tau(r \cdot A_r) > r + 2$. (For $r = 9$: if $3 \mid A_9$, additional factors of 3 only increase τ .)

Case $j \equiv 6 \pmod{11}$ (vacuous). Here $p = 420j - 1 \equiv 420 \cdot 6 - 1 = 2519 \equiv 0 \pmod{11}$. Since p is prime and $p \geq 419 > 11$, this is impossible. No valid p exists in this class.

This covers all nine residue classes $j \not\equiv 0, 7 \pmod{11}$, completing the proof. $\square$

Part C (computational verification). For the two remaining classes $j \equiv 0$ and $j \equiv 7 \pmod{11}$, the mod-11 algebraic argument does not yield a uniform witness from $k \leq 10$. (At $j = 6,970,590 \equiv 0 \pmod{11}$, for instance, every $k \leq 10$ gives $\tau(kA_k) \leq k + 2$; the smallest working witness is $k = 14$, giving $\tau(14A_{14}) = 64 > 16$.) A direct search over all proper divisors of 2520 does succeed:

Claim 1. *For every $j \in \{1, \dots, 10^7\}$ with $j \equiv 0$ or $7 \pmod{11}$, there exists $k \mid 2520$ with $1 \leq k < 2520$ such that $\tau(kA_k) > k + 2$.*

Verification procedure. Let D be the set of 47 proper divisors of 2520. For each j with $1 \leq j \leq 10^7$ and $j \equiv 0$ or $7 \pmod{11}$ (approximately 1.82×10^6 values), the following was executed: compute $A_k = (2520/k)j - 1$ and $\tau(kA_k)$ for each $k \in D$ in increasing order, halting at the first k with $\tau(kA_k) > k + 2$. The implementation used Python with the `sympy` library's `divisor_count` function, which computes the exact divisor count via integer factorisation. Every tested j produced a witness; no failures were found. The largest witness index required was $k = 14$, first occurring at $j = 6,970,590$ (where every $k \leq 10$ fails).

The numbers factored satisfy $kA_k < 2520j \leq 2520 \times 10^7 = 2.52 \times 10^{10}$. Since $\lfloor \sqrt{2.52 \times 10^{10}} \rfloor = 158,745$, complete factorisation of any such number requires only trial division up to 158,745; no probabilistic primality test is invoked. The computation is therefore deterministic and independently reproducible from the description above.

By Lemma 2, for every $j \not\equiv 0, 7 \pmod{11}$ (all $j \geq 1$), and by Claim 1 for $j \equiv 0, 7 \pmod{11}$ with $j \leq 10^7$, there exists k such that the witness $m = n - k$ satisfies $f(m) = (n - k) + \tau(kA_k) > (n - k) + (k + 2) = n + 2$.

The algebraic cases (Cases 1, 2, 3a, Sub-case 3b with $n \in \{30, 60\}$, and cases (i)–(v) with $p \not\equiv 419 \pmod{420}$) together with Lemma 2 and Claim 1 establish Theorem 1. The remaining infinite family ($j \equiv 0, 7 \pmod{11}$, $j > 10^7$) constitutes Conjecture 1's open part; see Remark 2.

$n = 24$ **satisfies the inequality.** The values $f(m) = m + \tau(m)$ for $1 \leq m \leq 23$ achieve maximum $f(20) = f(22) = 26 = 24 + 2$. Hence $\max_{m < 24} f(m) = 26 = 24 + 2$, confirming $n = 24$ is a solution and the largest one. $\square$

Remark 1. *The complete set of solutions is $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$. For each n from 1 to 10,000, one can compute $\max_{m < n} f(m)$ and verify directly that no $n > 24$ in that range satisfies the inequality.*

Proposition 1. *Let $\mathcal{E} = \{j \geq 1 : 420j - 1 \text{ prime and } 504j - 1 \text{ prime}\}$. Then $|\mathcal{E} \cap \{1, \dots, N\}| = O(N/(\log N)^2)$. In particular, the exceptional n -values have natural density zero.*

Proof. Set $L_1(j) = 420j - 1$ and $L_2(j) = 504j - 1$. For each prime p , let

$$\omega(p) = \#\{j \bmod p : p \mid L_1(j)L_2(j)\}$$

be the number of residue classes modulo p blocked by L_1L_2 .

Admissibility. We must show $\omega(p) < p$ for all primes p . For the small primes:

- $p \in \{2, 3, 7\}$: $p \mid 420$ and $p \mid 504$, so $L_1(j) \equiv -1$ and $L_2(j) \equiv -1 \pmod{p}$ for all j . Hence $\omega(p) = 0 < p$.
- $p = 5$: $5 \mid 420$ so $L_1(j) \equiv -1 \pmod{5}$ for all j ; no j blocks via L_1 . For L_2 : $504 \equiv 4 \pmod{5}$, so $L_2(j) \equiv 4j - 1$, which is $0 \pmod{5}$ only for $j \equiv 4 \pmod{5}$. Thus $\omega(5) = 1 < 5$.

For $p > 7$: $\gcd(420, p) = \gcd(504, p) = 1$, so L_1 and L_2 each have exactly one zero modulo p . If these zeros coincide, $\omega(p) = 1$; otherwise $\omega(p) = 2$. In either case $\omega(p) \leq 2 < p$.

Upper bound. Since $\omega(p) \leq 2$ for all p , the Euler product $\prod_p (1 - \omega(p)/p)(1 - 1/p)^{-2}$ converges to a finite nonzero constant (by comparison with the standard twin-prime product). The upper-bound Selberg sieve for two linear forms (Halberstam and Richert, *Sieve Methods* (1974), Theorem 5.1 with sieve dimension $\kappa = 2$; or Iwaniec and Kowalski, *Analytic Number Theory* (2004), Theorem 6.4) yields, for any $z \leq N$:

$$|\{j \leq N : L_1(j) \text{ and } L_2(j) \text{ both prime}\}| \ll \frac{N}{(\log z)^2} \prod_{p \leq z} \left(1 - \frac{\omega(p)}{p}\right) \left(1 - \frac{1}{p}\right)^{-2}.$$

Taking $z = N^{1/2}$ gives $|\mathcal{E} \cap \{1, \dots, N\}| \ll N/(\log N)^2$. $\square$

Remark 2. *The only unresolved cases are $n = 2520j$ where $j \equiv 0$ or $7 \pmod{11}$, $j > 10^7$, and both $420j - 1$ and $504j - 1$ are prime (equivalently: $p = 420j - 1$ prime and $(6p + 1)/5 = 504j - 1$ prime; see the note at the start of the hardest sub-case above). In every such case up to $j = 10^7$ a witness was found. Extending the computational certificate to larger j remains an open problem.*